MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
OPERATIONS DIVISION — CRUDE DISTILLATION UNIT (CDU-II)

STANDARD OPERATING PROCEDURE: HYDROCARBON FEED PUMP STARTUP & ISOLATION
DOC NO: **MRPL-SOP-CDU-0101** | REV: 04 | EFFECTIVE: 15-JAN-2026 | CLASSIFICATION: CONFIDENTIAL

1. PURPOSE & APPLICABILITY

This standard operating procedure mandates safety requirements, alignment steps, and operational guidelines for the startup, continuous operation, and emergency isolation of Crude Booster Pumps **P-101A/B** serving atmospheric distillation train CDU-II at MRPL Mangalore complex. Applicable to all plant operators and shift engineers.

2. SAFETY INTERLOCKS & OPERATIONAL THRESHOLDS

Prior to motor energization, field operations shall confirm that the DCS permissive interlock logic is SATISFIED and trip thresholds are active in accordance with Safety Integrity Level (SIL-2) allocation:

Tag	Parameter	Trip Setpoint	Alarm Setting	Interlock Logic
PSLL-101	Suction Pressure (P-101A)	1.80 bar (g)	2.05 bar (g)	Trip Motor instantly; prevent dry-run
PSHH-101	Discharge Pressure (P-101A)	18.50 bar (g)	17.20 bar (g)	Open bypass FCV-101 to 100%
TSH-101	DE Bearing Temperature	85.0 °C	75.0 °C	Alarm annunciator CDU console
TSHH-101	NDE Bearing Temperature	95.0 °C	85.0 °C	Trip Motor (2 out of 3 voting)
PDSL-101	Suction Basket Strainer ΔP	0.45 bar	0.30 bar	Switch over to standby strainer
FSLL-101	Minimum Flow Bypass	35.0 m³/h	42.0 m³/h	Open FCV-101; trip pump after 30s delay

3. PRE-STARTUP VERIFICATION CHECKLIST

• **Mechanical Seal Flush**: API Plan 53B barrier fluid reservoir pressure maintained at 3.5 bar (g), minimum 1.5 bar above seal chamber pressure.
• **Lube Oil System**: Check constant level oilers at 50% mark with ISO VG 46 turbine oil. Cooling water flow confirmed through bearing housings.
• **De-isolation Check**: Verify suction isolation valve MOV-101A is 100% locked open (LO). Confirm blind spectacle position is OPEN.
• **Priming and Venting**: Vent pump casing high point valve V-101-V until bubble-free hydrocarbon liquid appears into closed drain header.

4. STARTUP EXECUTION SEQUENCE

1. Request electrical clearance (Permit E-04) and ensure motor breaker racked IN.
2. Ensure minimum recirculation bypass valve **FCV-101** is 100% open to feed drum V-101.
3. Crack open discharge manual throttling valve MOV-101B to 10% position.
4. Depress local START pushbutton while monitoring motor inrush current (must decay from ~380A to normal 68A within 6.5s).
5. Observe discharge pressure gauge PI-1012 rise smoothly to shutoff head (~15.4 bar).
6. Slowly ramp discharge valve MOV-101B over 90 seconds to target flow of 145 m³/h while DCS places FCV-101 on automatic flow control.

5. EMERGENCY SHUTDOWN & ISOLATION

In the event of uncontrolled mechanical seal leakage, vapor lock, or fire alarm in battery limit: Press ESD-101 field button immediately. This executes: (a) Motor trip, (b) Immediate fail-close of suction valve MOV-101A and discharge valve MOV-101B, (c) Nitrogen purge sequence injection into pump stuffing box.